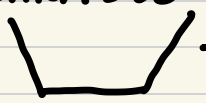
Week 4 Learning/Evaluations

- Gradient: What way I should change my weights to make mistake bigger
- Gradient Descent: going other way
- Anylatic gradient: gradient from back propogations
- Numerical gradient: change weight tiny bit and see what happens
  - ↳ Centered Gradient

$$\lim_{h \to 0} \frac{f(x+h) - f(x-h)}{2h}$$

- Relative Error: How big diff is btwn numerical/anylatical gradient
- Relu(0, x) → Activation function
  - ↳ Hard to get derivative so use sub-gradient
- Loss: How wrong prediction + How complicated solutions are
- Beginning of training
  - ↳ Give 20 examples
  - ↳ make it memorize examples so it is working properly
- Learning Rate: How big each step should be
- Validation accuracy on data set it hasn't encountered
- Low Training / Low Validation = the brain is too small
- Stochastic Gradient Descent (SGD)
  - New weight: x - learning rate · gradient
- Momentum SGD



→ velocity keeps helping it move

→ keep going forward

- Nestrov Momentum
  - ↳ Look ahead before letting momentum carry me
- Learning Rate Decay
  - Big steps than smaller steps
- Hessian Learning Rate
  - ↳ Tells curvature so you can take better steps
- Adaptive Learning Rate
  - ↳ Each weight has own gradient
- Adam Optimizer
  - ↳ Adjust weights based on curvature
- Hyperparameters arguments can't change